

AN AI ASSISTANT ON YOUR OWN MACHINE



github.com/aarora79/meru

Meru loads open-weight models into your computer's memory and keeps them there. You need no API key and pay for no tokens, so a question costs you electricity. No prompt leaves your machine unless you allow it.



मेरु

Meru is the cosmic mountain that the sun, moon and stars turn around. This assistant takes the name because it works the same way: it stays in one place, on your machine, and your notes, tools and daily routine turn around it.

~50 ms

For the meru client to start. The model is already loaded.

~2 GB

To download the default lite profile. Runs on 16 GB of RAM.

0

Code paths to a hosted model. The engine talks only to loopback.

1

Function, dispatch, that every tool and agent call passes through.

01 TWO PROGRAMS, BECAUSE BIG MODELS LOAD SLOWLY

merud + meru · Unix socket

MERUD · ALWAYS RUNNING

- 1 Keeps each model loaded in Ollama and warms every tier at startup.
- 2 Owns the store, the MCP and A2A connections, and the scheduler.
- 3 Runs the agent loop and emits metrics and traces.

MERU · THIN CLIENT

- 1 Connects to `~/meru/merud.sock`.
- 2 Sends your question and streams the answer back as it arrives.
- 3 Exits. It holds no model and no store logic.

The full profile's ~18 GB main model takes many seconds to load from disk, so the daemon holds it and the client stays thin.

```
$ meru "what did I change in the portfolio repo this week?"
$ meru chat # interactive terminal UI
$ meru index ~/notes ~/repos # build the local knowledge index
$ meru memory list # what it knows about you, in plain text
```

02 PICK THE PROFILE YOUR MEMORY CAN HOLD

lite is the default

TIER	JOB	LITE (DEFAULT)	FULL
fast	routing, query rewrites	MiniCPM5-2B, 1.6 GB	same
main	reasoning, choosing tools	same model as fast	qwen3.8:27b, ~18 GB at 4-bit
embed	vectors for the index	nomic-embed-text, 768 dims	qwen3-embedding:0.6b, 1024 dims
RAM	hardware	16 GB; a CPU is enough	Apple silicon, 32 GB+, or a 24 GB GPU

Code asks for a tier, and config.toml maps each tier to a model. Meru won't run 120B models: at 4-bit one needs about 60 GB and leaves even a 64 GB machine swapping.

03 FIVE STEPS IN A LOOP WE WROTE

no agent framework

1 · Route

The fast model picks: answer, search your files, call tools, or both.

2 · Context

Prompt, skills, memories and chunks, each under a token budget.

3 · Main

The main model streams its answer to you as it writes.

4 · Dispatch

Tool calls go through one function. Independent calls run at the same time.

5 · Repeat

Until the model answers without a tool call, or hits the cap of 8 rounds.

The allowlist, audit log, context budget and confirmation prompts all live in this loop, where you can read them. Ctrl-C cancels the whole turn.

04 WHY RUN IT YOURSELF

three reasons, in order

Privacy you can check

Your email, notes, money and calendar stay on your machine, out of anyone's training data.

No cost per question

Indexing ten years of notes or a daily brief costs nothing once you own the hardware.

It keeps working

No rate limits or price changes. The model on your disk in 2026 still runs in 2031.

05 THE RULES THE CODE KEEPS

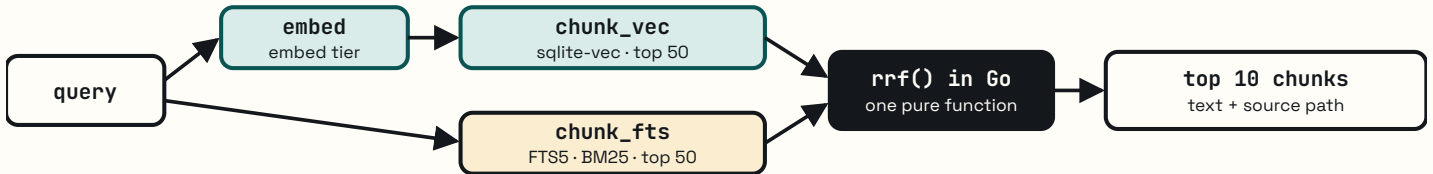
from the privacy boundary

- **No hosted-model code path.** The one model client talks to Ollama and refuses any URL off loopback.
- **Deny by default.** MCP tools and A2A skills start switched off; you allow them one at a time.
- **Every external action logged.** Review them all with `meru log`.
- **No telemetry.** No crash reports, and no update checks.
- **Loopback unless you say so.** A remote server or agent needs `network = true` in its entry.
- ! **No sandbox.** MCP servers run with your permissions, so add only ones you trust.

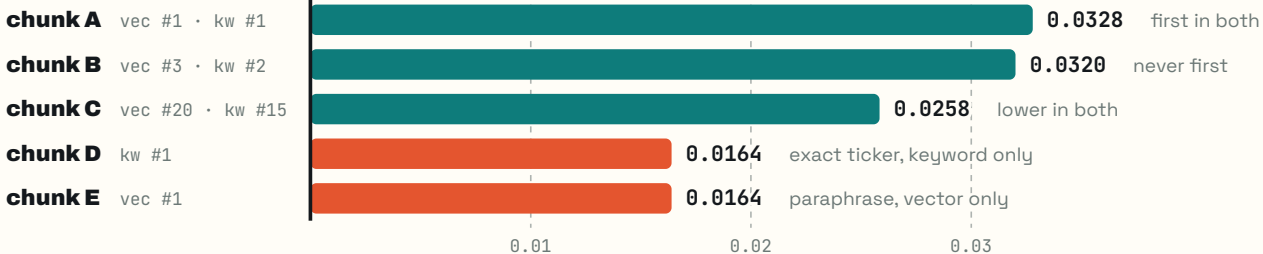
06 TWO SEARCHES, MERGED BY RANK

BM25 + sqlite-vec + reciprocal-rank fusion

Vector search misses exact strings such as ticker symbols and error codes. BM25 keyword ranking catches those strings and misses paraphrase. Meru runs both inside SQLite and merges the two ranked lists in Go.



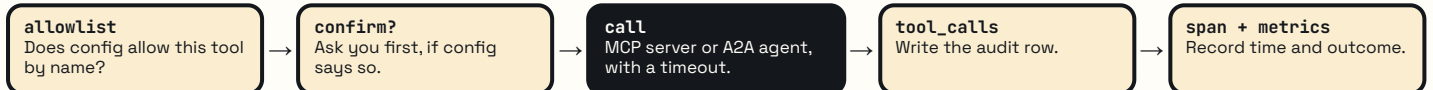
RRF SCORE, $K = 60 \cdot \text{RANKS INVENTED, SCORES COMPUTED}$



Fusion ignores the two lists' scales and uses position: $\text{score} = \sum 1/(60 + \text{rank})$. B ranks third and second, and scores almost twice as high as D or E.

07 EVERY TOOL AND AGENT CALL PASSES ONE FUNCTION

dispatch · deny by default



No other code path reaches a server or an agent. A server that offers 40 tools gives the model none until you allow them by name. Any MCP server you already run works with no new code. Each allowed A2A agent skill becomes one more tool, `a2a.<agent>.<skill>`, with the same checks.

```
[[mcp.servers]]
name = "obsidian"
command = "."
allow = ["read", "search"] # stdio child # the rest: off
confirm = [] # ask me each call
```

08 DELETE MERU.DB AND MERU REBUILDS IT

files are the source of truth

FILES · THE TRUTH

- Your folders: notes, docs, PDFs, repos.
- `~/.meru/sessions/`: one JSONL file per session, append-only.
- `config.toml` and `skills/`, markdown you can edit.

~/.MERU/MERU.DB · THE INDEX

- chunks, `chunk_fts` and `chunk_vec`, from your folders.
- messages and `message_fts`, replayed from the sessions.
- `tool_calls`: the audit log, one readable row per call, with its trace ID.

One SQLite file through `ncruces/go-sqlite3`, with `sqlite-vec` built in, so the build needs no C compiler and no second database.

09 TRACES AND METRICS THAT STAY ON YOUR MACHINE

OpenTelemetry · loopback only

- Export stays off until you set `observability.otlp_endpoint`.
- `merud` refuses to start if that endpoint isn't loopback.
- Spans leave out prompt text unless you set `capture_content = true`.
- The `grafana/otel-lgtm` stack binds to `127.0.0.1`, with reporting off.

```
meru.turn route, iterations, outcome
├─ meru.route fast-tier call
├─ meru.retrieve vector, fts, fusion
├─ gen_ai.chat main
├─ mcp.tool_call obsidian.search
├─ gen_ai.chat main final answer
```

10 SHIPPING ORDER

each milestone useful by itself

v0.1 IT TALKS

A cold meru "hello" streams a first token in under 1s.

v0.2 IT KNOWS YOUR FILES

It answers from a local note and cites the file.

v0.3 IT CAN DO THINGS

It answers by calling an MCP server you already run.

v0.4 IT'S YOURS

It recalls something from last week without a reminder.

v0.5 IT ACTS UNPROMPTED

A morning brief arrives without you asking.

Pre-alpha, design phase. This poster describes the design in `ARCHITECTURE.md`, which the code will follow. The 50 ms start and the first-token target are design goals that no one has measured yet. Model sizes come from the model publishers. Panel 06 uses made-up ranks to show the formula; the scores follow from it.



READ THE DESIGN, FOLLOW THE BUILD

github.com/aarora79/meru — `README.md`, `ARCHITECTURE.md`, `ROADMAP.md`

Amit Arora · MIT license · macOS, Linux, Windows · Ollama on loopback · Go binaries you can copy to another machine and run.